

Inferential statistics

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About me



Background:

- Senior Lecturer, Department of Community Medicine, UniSZA
- Education:
 - PhD, Department of Community Medicine, USM
 - MSc (Medical Statistics) from USM, 2019
 - MBBCh from Al-Azhar University, 2015

Interest:

- Medical statistics, machine learning application in medical sciences
- Application of R (and Python) in medical data

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Material

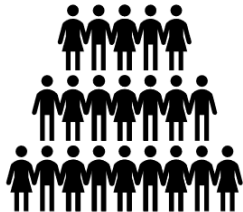
- All materials can be downloaded from
https://github.com/tengku-hanis/mahtas_workshop.git



Introduction

- Statistics is the science of learning from data
- Statistics provide framework for collecting, organising, analysing and interpreting information from the data to separate a meaningful signals (insight) from random noise
- The field can be divided into:
 - Descriptive statistics - relate to the process of summarising and organising data at hand (e.g., mean, standard deviation, median, and interquartile range)
 - Inferential statistics - relate to the process of making a prediction, test hypotheses, or draw conclusion about a larger that is impossible to measure

Population



Sampling



Sample



Inferential Statistics



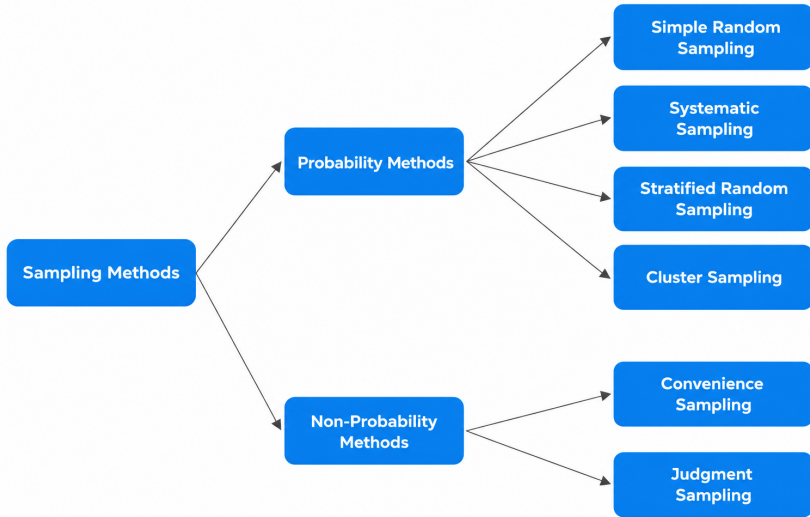
sampling error may occur

Descriptive Statistics

- Basically, inferential statistics exists because scientists realise that testing or measuring everyone in the population is impossible or too costly
- How we draw the sample from a population is known as sampling
- Sampling can be divided into two:
 - Probability sampling - each person in the population has the a known non-zero chance to be selected as a sample
 - Non-probability sampling - not every person in the population has the same chance to be selected as a sample

Probability sampling

- Probability sampling is preferred as the sample derived from this method is more likely to be representative of the population
- The sample derived from the probability sampling is known as a random sample
- It's not uncommon to do inferential statistics on a sample from the non-probability sampling process, however:
 - The derived inferential statistics has limited generalisability - the researcher should acknowledge this limitation



- Because we are making inferences about an entire population based on a smaller sample, there is inherent uncertainty
- This uncertainty is known as sampling error - the natural variation or difference between a sample statistic and the true population parameter
- In statistics, we quantify and reflect this uncertainty using a 95% confidence interval, which provides a range of values where we are 95% confident the true population parameter lies

- When we take a sample and calculate its mean, that sample mean has some sampling error
- But if we were to take hundreds of samples and plot all their means, those means would form a perfect bell curve known as a normal distribution - this process is describe as a central limit theorem
- The normal distribution has a mathematical properties that allow us to measure our uncertainty (i.e. the 95% confidence interval)

95% Confidence interval

$$95\% \text{ CI} = \text{Sample Mean} \pm (1.96 \times \text{Standard Error})$$

- Sample Mean (our best guess): Because we cannot measure the entire population, we start with the mean of our sample
- Standard Error (the sampling error):
 - This is the direct mathematical measurement of sampling error
 - It tells us how much we expect our sample mean to fluctuate from the true population mean just by random chance
- 1.96 (the 95% rule): Because we want to be 95% confident, the rules of the normal distribution tell us we need to multiply our error by roughly 2 (exactly 1.96)

- The $\pm$ Portion (the buffer zone):
 - The entire second half of the formula ($1.96 \times \text{Standard Error}$) is called the Margin of Error this represents our total uncertainty
 - We add and subtract this uncertainty from our sample mean to build our final interval